

The Yukawa Mechanism as a Consequence of $\tilde{T} \cdot m = 1$

Mass-proportional coupling as a derivation, not an assumption

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Correction notes

Corrections from the calculation review (1 Oct. 2026). The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- Yukawa coupling: only the exponents p_i follow from the topology; the quark prefactors r_i are open (previously "no free parameter" **[B]** and "0 free parameters").
- Rational ratios: $M_W^2/M_Z^2 = 7/9$ agrees to 0.13 % (previously 0.06 %, which applies to M_W itself).

Abstract. In the Standard Model the Yukawa coupling $y_i = m_i/v$ is an empirical assumption: it is chosen so that the observed masses emerge after electroweak symmetry breaking. This document shows that the same coupling follows from the FFGFT ground relation $\tilde{T} \cdot m = 1$ once the electroweak scale v is allowed to depend on position. The mass-proportional coupling of all fermions to the Higgs is then not an assumption but the fluctuating form of $\tilde{T} \cdot m = 1$. The Higgs particle $h(x)$ is the vibrational quantum of the lattice scale — not an independent scalar field with an unknown potential. Its mass is the residual stiffness of the lattice: $M_W^2 + M_Z^2 + m_h^2 = v^2/2$ (trace rule, 0.45 %). Together with $\sin^2 \theta_W = 2/9$ and $m_h/M_Z = 11/8$, M_Z , M_W and m_h follow from v alone to within 0.2–0.3 %.

Notation

- [B]** algebraically proven
- [K]** numerically confirmed
- [S]** structurally plausible, derivation open
- [X]** tested and negative

Symbol	Meaning
$\tilde{T}_i$	proper-time scaling of mode i
m_i	rest mass of particle i
v	electroweak vacuum scale (246.22 GeV)
$h(x)$	physical Higgs field (fluctuation of v)
y_i	Yukawa coupling (SM definition: $m_i = y_i v$)
r_i, p_i	prefactor and exponent of the FFGFT mass formula
ξ	FFGFT scaling parameter (4/30000)
$\psi_i, \bar{\psi}_i$	fermion field of particle i and its Dirac adjoint
$\mathcal{L}$	Lagrangian density
H	complex lattice amplitude (Higgs field), $H = (v + h)/\sqrt{2}$
$n_{\phi,i}, n_{\theta,i}$	winding numbers on the T^4 torus
n_c	colour multiplicity
$d_{\text{akt},i}$	number of active torus dimensions (Doc. 189)
M_W, M_Z	masses of the W and Z bosons (pole masses)
m_h, m_t	mass of the Higgs particle, of the top quark
m_s	mass of the strange quark
$W^\pm$	charged gauge boson; one complex mode, $W^\pm = (W^1 \mp iW^2)/\sqrt{2}$
W^3, B	neutral gauge fields before mixing ($SU(2), U(1)$)
γ	photon
θ_W	Weinberg angle, on-shell $\sin^2 \theta_W = 1 - M_W^2/M_Z^2$
g, g'	gauge couplings of $SU(2)$ and $U(1)$
λ	Higgs self-coupling, $m_h^2 = 2\lambda v^2$
y_t	Yukawa coupling of the top quark
G_F	Fermi constant
τ_μ	muon lifetime (the only measured quantity behind v)
ϕ	golden ratio $(1 + \sqrt{5})/2$
Q	Koide quotient $\sum m_i / (\sum \sqrt{m_i})^2$
α, α_s	fine-structure constant, strong coupling
Λ_{QCD}	confinement scale of the strong interaction
R_{proton}	charge radius of the proton
$\theta_{12}, \theta_{13}, \theta_{23}$	neutrino mixing angles
δ_{CP}	CP-violating phase
K_{frak}	fractal correction factor $1 - 100\xi$ (Doc. 133)
$\text{GF}(q), \text{GF}(q)^*$	Galois field with q elements, its multiplicative group
A_5	icosahedral group (60 elements)

1 The Ground Relation and Its SM Notation

FFGFT sets for every fermion mode i :

$$\tilde{T}_i \cdot m_i = 1, \quad m_i = r_i \xi^{p_i} v. \quad (1)$$

The Yukawa coupling of the Standard Model is defined by $m_i = y_i v$, hence:

$$y_i = r_i \xi^{p_i}. \quad (2)$$

This is not a matching but a direct rewriting **[B]**. Here (Doc. 189):

$$r_i = \frac{n_{\phi,i}^2}{n_{\theta,i} n_c}, \quad p_i = \frac{d_{\text{akt},i}}{3}, \quad (3)$$

where $n_{\phi,i}$, $n_{\theta,i}$ are the winding numbers on the T^4 torus and n_c is the colour multiplicity. The exponents p_i follow from the topology (Docs. 189/358) **[K]**; the prefactors r_i follow from the winding numbers for the leptons, while for the quarks their derivation is open **[S]**. *(Corrected on 1 Oct. 2026: previously "The Yukawa coupling is thus fully determined by the torus topology — it has no free parameter **[B]**" — the status table in §7 lists the quark r_i as "derivation open", Doc. 372 calls them back-computed from measured values, and Doc. 190, R129 leaves open whether they are geometrically enforced.)*

2 Derivation of the Yukawa Vertex from $\tilde{T} \cdot m = 1$

The vacuum case

In the vacuum $v = \text{const.}$ Eq. (1) gives $\tilde{T}_i = 1/m_i$ — every mode has a fixed proper-time scaling. The Lagrangian mass term reads:

$$\mathcal{L} \supset -m_i \bar{\psi}_i \psi_i. \quad (4)$$

Fluctuating lattice scale

Now allow v to fluctuate with position:

$$v(x) = v + h(x), \quad |h(x)| \ll v. \quad (5)$$

The ground relation $\tilde{T}_i \cdot m_i = 1$ must continue to hold — not as a global normalisation but as a local bond between proper time and mass. This means:

$$m_i(x) = \frac{1}{\tilde{T}_i} = r_i \xi^{p_i} v(x) = r_i \xi^{p_i} (v + h(x)) = m_i^{(0)} \left(1 + \frac{h(x)}{v} \right). \quad (6)$$

The coupling term

Inserting into the mass term gives **[B]**:

$$\mathcal{L} \supset -m_i(x) \bar{\psi}_i \psi_i = -m_i^{(0)} \bar{\psi}_i \psi_i - \frac{m_i^{(0)}}{v} \bar{\psi}_i \psi_i h(x). \quad (7)$$

The second term is exactly the Yukawa vertex of the Standard Model with $y_i = m_i^{(0)}/v$. Here it does not follow from an assumption about the coupling structure but from the single FFGFT relation $\tilde{T} \cdot m = 1$ under spatial fluctuations of v .

3 Mass-Proportional Coupling as a Consequence

Eq. (7) holds for *all* fermions i *simultaneously*, with the same v and the same $h(x)$. This means:

[B] The mass-proportional coupling $y_i = m_i/v$ of all fermions to the Higgs boson is a necessary consequence of $\tilde{T} \cdot m = 1$ under spatial fluctuation of the lattice scale. It is not a free assumption of the model.

This is the content of the Yukawa mechanism in FFGFT language: not “particles couple to a scalar field” but “mass is a local property of the lattice, and $h(x)$ is the oscillation of this property”.

4 Interpretation of the Higgs Particle

In this derivation the field $h(x)$ is the position-dependent deviation of the lattice scale v from its vacuum value. The physical Higgs particle — the resonance at 125.2 GeV — is then the vibrational quantum of this deviation.

This differs conceptually from the Standard Model:

	Standard Model	FFGFT
Origin of mass	Yukawa coupling (postulated)	$\tilde{T} \cdot m = 1$ (ground relation)
Higgs field	independent scalar field	fluctuation of $v(x)$
Coupling $y_i = m_i/v$	empirical parameter	derived from Eq. (6)
Number of free parameters	n_{fermions}	0 in the lepton sector; quark r_i so far from measured values
Higgs potential $V(H)$	Mexican hat (postulated)	residual stiffness of the lattice (§6)
Higgs mass m_h	free parameter	trace rule (§6) [K]

(Corrected on 1 Oct. 2026: row “Number of free parameters”, FFGFT column, previously 0 – holds only in the lepton sector; the quark r_i are so far determined from measured values, see §7.)

The derivation so far shows that $h(x)$ *must* exist and *must* couple with m_i/v . The restoring force of the oscillation — the potential $V(h)$ and hence the Higgs mass — is treated in §5 and §6.

5 Higgs Mass from the Galois Chain [K]

Potential, Higgs mass and oscillation frequency are a single problem. This section shows that a numerically sharp candidate exists that is built entirely from Galois structures of the corpus.

The 11/8 chain. The three quantities M_Z , m_h , m_t satisfy:

$$M_Z : m_h : m_t = 1 : \frac{11}{8} : \left(\frac{11}{8}\right)^2, \quad m_h = M_Z \cdot \frac{11}{8}, \quad m_t = M_Z \cdot \left(\frac{11}{8}\right)^2. \quad (8)$$

Numerically **[K]**:

Quantity	measured	FFGFT (Eq. (8))	deviation
M_W	80.369 GeV	$M_Z \sqrt{7/9} = 80.420$ GeV	0.06 %
M_Z	91.188 GeV	input	—
m_h	125.200 GeV	$M_Z \cdot 11/8 = 125.383$ GeV	0.15 %
m_t	172.570 GeV	$M_Z \cdot (11/8)^2 = 172.402$ GeV	0.10 %

All four heavy bosons and fermions follow from two Galois numbers plus v .

Galois origin of the numbers. Both factors are field quantities of the corpus: $8 = |\text{GF}(9)^*| = 3^2 - 1$ (Doc. 336) and $11 \in \text{orbit}\{7, 11, 21\}$ under $x \mapsto 3x \bmod 26$ in $\text{GF}(27)^*$ (Doc. 340, the same structure as Δm_{atm}^2). The Weinberg factor $\sqrt{7/9}$ comes from the A_5 matrix element $2/9$ (Doc. 372). The chain thus links three different Galois structures into one consistent numerical picture **[K]**.

Classification. Eq. (8) is a numerically confirmed relation; the Galois origin of the numbers 11 and 8 is a reading **[S]**. A derivation of the Higgs mass that does without these numbers is provided by the trace rule in §6; together the two fix the entire electroweak boson spectrum from v alone.

6 The Trace Rule: Higgs Mass as Residual Stiffness of the Lattice

Picture. Coupled resonant circuits detune each other: the coupling pushes the eigenfrequencies apart. What is conserved in the process is the sum of the squared frequencies — the trace of the stiffness matrix. In this picture the Higgs mass is not a resonance of its own with its own potential but the share of the lattice stiffness left over by the gauge bosons.

The rule.

$$M_W^2 + M_Z^2 + m_h^2 = |\langle H \rangle|^2 = \frac{v^2}{2}, \quad m_h^2 = \frac{v^2}{2} - M_W^2 - M_Z^2. \quad (9)$$

The four building blocks.

1. **Conservation under coupling [B].** The sum of the eigenvalues of a symmetric matrix is its trace, independent of the basis. The electroweak mixing is a pure rotation by θ_W : before mixing, W^3 and B carry the stiffnesses $g^2 v^2/4$ and $g'^2 v^2/4$, afterwards the photon (0) and the Z (M_Z^2) — the sum is identical, the mixing matrix orthogonal.
2. **No intrinsic stiffness of the coupling [B].** A mechanical coupling spring is a component of its own and adds stiffness to the diagonal. The vacuum is empty; the coupling can only redistribute. That is exactly the property of a rotation (point 1).
3. **Counting once per mode.** W^+ and W^- are one complex mode with two senses of circulation: in the corpus the antiparticle is the same field with reversed sign (Doc. 042, 053), and the trace runs over m^2 , where the sign drops out. Consistently, Koide counts three lepton masses, not six.
4. **Budget = $v^2/2$ [B].** The Higgs *field* is the ground lattice itself; the total stiffness belongs to it and is the squared modulus of its vacuum amplitude. Since the amplitude is complex (modulus and phase), the canonical normalisation of the real oscillation h requires the factor $1/\sqrt{2}$: $H = (v + h)/\sqrt{2}$, hence $|\langle H \rangle|^2 = v^2/2$. It is the same $\sqrt{2}$ as in $W^\pm = (W^1 \mp iW^2)/\sqrt{2}$ — the argument that counts W once also fixes the factor $1/2$.

The Higgs *particle* h is the eigen-oscillation of the lattice; its stiffness m_h^2 is the remainder in Eq. (9). The Mexican-hat potential of the Standard Model is therefore not postulated: its curvature at the minimum is this residual stiffness.

Numbers [K]. With $M_W^2 = \frac{7}{9} M_Z^2$ (Doc. 372) it follows that $m_h = \sqrt{v^2/2 - \frac{16}{9} M_Z^2} = 124.6 \text{ GeV}$ (measured 125.2 GeV , -0.47%). The balance itself holds to 0.45% .

Boson spectrum from ν alone [K]. Combining Eq. (9) with $\sin^2 \theta_W = 2/9$ and the chain $m_h = \frac{11}{8} M_Z$ (§5) leaves

$$M_Z^2 \left(\frac{7}{9} + 1 + \frac{121}{64} \right) = \frac{\nu^2}{2} \implies \frac{M_Z^2}{\nu^2} = \frac{288}{2113},$$

and all three boson masses follow from ν without any measured mass as input:

	FFGFT	measured	deviation
M_Z	90.90 GeV	91.19 GeV	−0.31 %
M_W	80.17 GeV	80.37 GeV	−0.25 %
m_h	124.99 GeV	125.20 GeV	−0.17 %

As couplings: $g = 0.6512$ (on-shell $2M_W/\nu = 0.6528$), $\lambda = 0.1288$ (from m_h : 0.1293); the Higgs self-coupling is fixed by $4g^2/7 + 2\lambda = 1/2$ and is no longer a free parameter.

Accuracy. All three values lie 0.2–0.3 % low, in the same direction. Eq. (9) is a tree-level relation; the electroweak radiative corrections are of the order of a few per cent. Without a loop calculation no better agreement than the one reached here can be expected. The choice of ν shifts the result: with the SM definition 246.22 GeV the balance holds to +0.45 %, with the FFGFT values from Doc. 006/R104 (248.9 and 245.6 GeV) to −1.7 and +1.0 % respectively. Which ν is the right one is the question registered in R104/R112 (convention-free only via τ_μ), not a gap in the trace rule.

Fermion side: the trace rule applies to lattice modes only. The obvious extension “the top carries the same budget”, $m_t^2 = \nu^2/2$ ($y_t = 1$), is incompatible with the chain: together with Eq. (9) it requires $m_t/M_Z = \sqrt{2113/576} = 1.915$, while the measured value is 1.8925 (+1.2 %) and $(11/8)^2 = 1.8906$ agrees to 0.10 %. The reading $y_t = 1$ is discarded [X]: it has no justification beyond the analogy, and it is the worse one. In addition, the top mass itself is scheme-dependent — pole mass 172.6 GeV, $\overline{\text{MS}}$ mass 162.5 GeV, a spread of 5.8 %, corresponding to $y_t = 0.99$ and 0.93. Quarks are not free asymptotic modes (Doc. 372 §4); the trace rule applies to the eigen-oscillations of the lattice (W, Z, h), not to confined fermions. For the top mass the chain $m_t = (11/8)^2 M_Z$ applies [K].

Rational ratios versus ν -dependent relations. The accuracy of the relations is ordered by whether they contain ν :

Relation	contains	deviation
$m_t/M_Z = (11/8)^2$	pole masses only	0.10 %
$M_W^2/M_Z^2 = 7/9$	pole masses only	0.13 %
$m_h/M_Z = 11/8$	pole masses only	0.15 %
$M_W^2 + M_Z^2 + m_h^2 = \nu^2/2$	ν	0.45 %
$m_t^2 = \nu^2/2$ (discarded)	ν	1.2 %

(Corrected on 1 Oct. 2026: previously $M_W^2/M_Z^2 = 7/9$ with 0.06 % in first place – $(M_W/M_Z)^2 = 0.77679$ against $7/9 = 0.77778$ gives −0.13 %; the 0.06 % apply to M_W itself ($M_Z\sqrt{7/9} = 80.420$ against 80.369 GeV). Order adjusted accordingly.) Ratios of measured pole masses are convention-free; this is where the rational Galois numbers agree best — as is to be expected, since they are exact numbers, not parameters. ν , by contrast, is not a measurement but the SM definition $(\sqrt{2} G_F)^{-1/2}$, whose irrational factors stem from the SM formulation and which carries a theory correction of −0.44 % from muon decay (Doc. 351, R112). Every relation containing ν inherits these conventions. The residual of the trace rule (0.45 %) is of the size of this correction; that it stems from the definition of ν and not from the rule is a testable conjecture [S].

7 Inventory of the Particle Zoo (as of 21 September 2026)

The following overview summarises what the corpus, after Doc. 372 and this document, delivers for each sector. In each case the best derivation is decisive; older, superseded formulas are recorded in the register (Doc. 190).

Sector	Quantity	Source	Status
Charged	mass ratios $e : \mu : \tau$ from the matrix element 2/9	292, 293, 367	[B]/[K]
leptons	ϕ skeleton of the weights, $4/3 = 2Q$	368, 371	[B]
Neutrinos	masses from the GF(27)* orbits	340	[K]
	ν_3 Dirac, $\nu_{1,2}$ Majorana	343	[B]
	$\sin^2 \theta_{13} = (25\xi)^{2/3}$	340	[K]
	$\sin^2 \theta_{23} \in \{4/9, 5/9\}$, $ \sin^2 \theta_{23} - \frac{1}{2} = \frac{1}{18}$	372	[K]
	θ_{12} (order of magnitude only), δ_{CP}	340	[S]
	absolute scale m_ν (ansatz)	007	[S]
Gauge bosons	photon and eight gluons massless	320, 321, 339	[B]
	$M_W^2/M_Z^2 = 7/9$ ($\sin^2 \theta_W = 2/9$ on-shell)	372	[K]
	$M_Z^2/v^2 = 288/2113$ (M_Z from ν alone)	373	[K]
Higgs	Yukawa coupling $y_i = m_i/v$ from $\tilde{T} \cdot m = 1$	373	[B]
	$m_h = \frac{11}{8} M_Z$	373	[K]
	trace rule $M_W^2 + M_Z^2 + m_h^2 = v^2/2$, self-coupling fixed	373	[K]
Top	$m_t = (\frac{11}{8})^2 M_Z$	373	[K]
Other quarks	exponents $p_i = d_{akt}/3$	189	[K]
	prefactors r_i in the Galois grid; derivation open	006, 358	[B]/[S]
	unrolling as for leptons ($Q = 2/3$)	372	[X]
Hadrons	confinement as a topological boundary condition	319	[S]
	Gell-Mann–Okubo splitting consistent with m_s	372	[K]
	R_{proton} structurally classified	372	[S]
	Λ_{QCD} from the ξ ladder	372	[X]
Couplings	α from Galois orders	338	[K]
	α_s via the Galois running	160, R78	[K]
	Cabibbo angle	372	[S]
Scale	ν from the lattice	006, R104	[K]
	convention-free test only via τ_μ	351, R112	[S]

Ordering principle. Precise are the free modes — the charged leptons — and the oscillations of the lattice itself (W , Z , h). Blurred is everything confined: quarks and hadrons, whose masses depend on the scheme. And the exact Galois ratios between pole masses agree better than any relation that contains the SM quantity ν (§6).

Open. θ_{12} and δ_{CP} ; the quark prefactors r_i apart from the top; Λ_{QCD} ; the Cabibbo angle; the geometric derivation of the hadronic factor in Doc. 160 (R123); the convention-free determination of ν via the muon lifetime (R112).

8 Result

1. The Yukawa coupling $y_i = r_i \xi^{p_i}$: exponents p_i from the topology (Docs. 189/358) **[K]**; prefactors r_i for leptons from winding numbers, for quarks derivation open **[S]**. *(Corrected on 1 Oct. 2026: previously "is a direct rewriting of the torus topology — not a free assumption **[B]**" — see the correction in the section on the ground relation.)*
2. The Yukawa vertex $-(m_i/v) \bar{\psi}_i h \psi_i$ follows from $\tilde{T} \cdot m = 1$ under $v \rightarrow v + h(x)$ for all fermions simultaneously **[B]**.
3. The mass-proportional coupling (the fingerprint of the Higgs discovery) is a consequence, not a parameter **[B]**.
4. $h(x)$ is the vibrational quantum of the lattice scale. The Higgs potential is not a separate postulate: its curvature at the minimum is the residual stiffness of the lattice (§6).
5. **Higgs mass candidate **[K]****: $m_h = M_Z \cdot 11/8 = 125.383 \text{ GeV}$ to 0.15 %; $m_t = M_Z \cdot (11/8)^2$ to 0.10 %. Both factors — $11 \in \text{GF}(27)^*$, $8 = |\text{GF}(9)^*|$ — are Galois quantities of the corpus; their origin is a reading **[S]**.
6. **Trace rule**: $M_W^2 + M_Z^2 + m_h^2 = v^2/2$ to 0.45 % **[K]**. Conservation under coupling, absence of intrinsic stiffness, counting and the factor 1/2 are justified **[B]**; $m_h = \sqrt{v^2/2 - \frac{16}{9} M_Z^2}$.
7. **Boson spectrum from v **[K]****: $M_Z^2/v^2 = 288/2113$; M_Z , M_W , m_h to 0.17–0.31 %, at tree level.
8. Inventory of the entire particle zoo in §7.
9. Top: the trace rule applies to lattice modes only; $y_t = 1$ discarded **[X]**, $m_t = (11/8)^2 M_Z$ **[K]**.
10. Accuracy is ordered by convention dependence: rational pole-mass ratios 0.06–0.15 %, relations with v 0.45 %. Open **[S]**: whether the residual of the trace rule stems from the definition of v (R104/R112).

Check script: 2/python/Dok373_Skripte/pruef_373_yukawa.py

Register (Doc. 190): R122 updated (Higgs mass: candidates available).

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